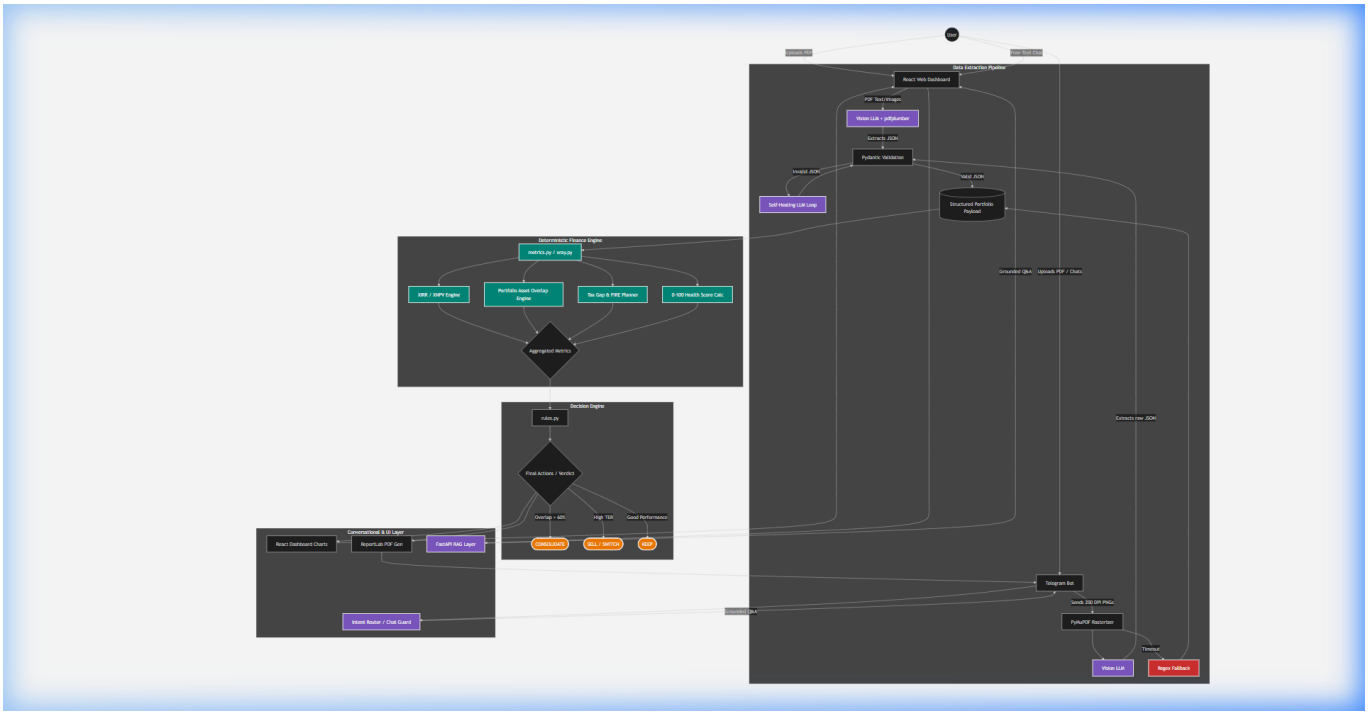
Complete System Architecture (FinMentor AI & ArthaScan)

> [!IMPORTANT] This system is built on **"Zero-Hallucination Finance,"** isolating mathematical logic from generative AI to ensure 100% accurate financial advice.

🏗 Unified Flow



🛠 Component Breakdown (1-Page Summary)

1. Multi-Channel Extraction (Vision-First)

Standard text extraction often fails on complex statements. We use **Vision LLMs** to "read" document images:

- **Web Dashboard:** Combines `pdfplumber` with Vision-capable LLMs for structural parsing.
- **Telegram Bot:** Rasterizes PDFs to 200 DPI PNGs via `PyMuPDF` for high-accuracy image-to-JSON extraction.
- **Error Handling:** Features a **Self-Healing Loop** (Pydantic re-prompts for JSON repairs) and **Regex Fallbacks** for resilient data capture.

2. Deterministic Financial Engine (The Sandbox)

AI is banned from calculations. A static Python engine processes the validated JSON payload:

- **XIRR Engine:** Uses `XNPV` binary-search for true annualized returns.
- **Duplication Engine:** Intersects fund holdings to find hidden asset overlap.
- **Wealth Bleed:** Calculates 10-year fee erosion vs. index baselines.

3. Agent Roles & Decisions

- **Extraction Agent:** Converts messy PDFs into structured "Financial Truth" dictionaries.
- **Decision Engine:** A rigid heuristic tree (rules.py) that issues `SELL`, `SWITCH`, or `CONSOLIDATE` commands based on math—not probability.
- **Presentation Agent:** Translates JSON findings into fluid conversational English/Hinglish (Chat Guards prevent hallucinations).

4. Tool Integrations

Interface	Tech Stack	Primary Tools
Backend	FastAPI / Python	pyxirr, numpy, pydantic
Frontend	React 18 / Recharts	Animated ScoreRings, Heatmaps
Bot	Telegram Bot API	ReportLab (PDF Gen), Cache